Historical Accuracy of Rolling Focus IPCA Forecasts

Abstract

- Main question: how do Focus-based IPCA forecast errors vary across rolling 3-, 6-, and 12-month horizons?
- Secondary questions:
  - How is 12-month forecaster disagreement related to error?
  - What is the performance of these forecast errors compared with an order one autoregressive rule?
- Main finding: cumulative mean absolute error (MAE) is 0.434, 0.953, and 1.794 percentage points, respectively, across 279 matched forecast months.
- State the main interpretation and its horizon-scale limitation.

1 Introduction

- Motivate the comparison of past Focus expectations with realized Brazilian IPCA.
- State the two research questions and explain the rolling forecast windows.

2 Methods

2.1 Data

The Central Bank of Brazil publishes a weekly suvey on the expectations of economic indicators from financial market players, called the Focus Report. From this survey, we can recorver the estimate for the consumer inflation rate of Brazil, the IPCA, for following 3, 6 and 12 months. On the other hand, the Brazilian Institute for Statistcs (IBGE) publishes the monthly realized value for the IPCA. This study will use data from both of these sources to calculate statistics and produce simulations regarding the error between expected and realized value of inflation in Brazil. We use 279 forecast dates, one per month from November 2001 through January 2025.

2.2 Forecasts and error measures

The core measurement for this work will be the signed error for the inflation expectations, defined as the forecast minus the realized inflation. From this, we will also relport absolute-error measures, including mean absolute errors (MAE), quantiles, and empirical cumulative distribution functions (ECDFs). We will also explore if the signed error is related to the standart deviation of the prediction.

2.3 Annual target check: direction and closeness

- Treat the 25 calendar-year forecasts in the annual notebook as a separate sample from the rolling monthly panel.

- Define relative error as $|\hat{\pi}_y - \pi_y|/|\pi_y|$, where $\hat{\pi}_y$ is forecast IPCA and π_y is realized IPCA. Call a forecast "90% close" when the error is at most 0.10 (within 90–110% of the realized value), and "95% close" when it is at most 0.05. This treats under- and overestimation symmetrically.
- Show the signed-error distribution around zero and an ECDF of relative error with markers at 5% and 10%.

2.4 Simulation

- Use 10,000 paired circular moving-block bootstrap draws with 12-month blocks and seed `20261102`.
- Report percentile 95% intervals, convergence at 1,000/5,000/10,000 draws, and sensitivity to 6- and 24-month blocks.

3 Results

3.1 Main finding and descriptive statistics

Constructed horizon	MAE (p.p.)	Mean signed error (p.p.)
3 months	0.434	-0.165
6 months	0.953	-0.436
12 months	1.794	-1.033

- The 12-minus-3-month MAE difference is 1.360 p.p.; its 12-month-block bootstrap 95% interval is 0.788–2.089 p.p.
- Present the ECDFs, error distribution shapes, under/overestimation frequencies, and the other paired horizon contrasts.

3.2 Published 12-month benchmark and disagreement

- Published 12-month MAE: 1.809 p.p.; mean absolute gap from the constructed 12-month forecast: 0.049 p.p.
- Spearman correlation between published forecast dispersion and subsequent absolute error: $\rho = 0.245$ (bootstrap 95% interval: 0.046–0.413).
- Show the forecast comparison, disagreement scatter plot, bootstrap distribution, and convergence diagnostics.

3.3 Annual target check: descriptive results

- Of 25 annual forecasts, 16 underestimated realized IPCA and 9 overestimated it; mean signed error was -0.984 p.p.
- Seven forecasts (28%) were within 10% of realized inflation; one (4%) was within 5%. Show these counts on the relative-error ECDF.

4 Discussion

- Interpret the results while accounting for the different scales of cumulative 3-, 6-, and 12-month inflation.

- Keep the annual 25-year check distinct from the 279-month rolling analysis. The 90% and 95% closeness thresholds are descriptive tolerances, not confidence levels.
- Discuss overlapping windows, possible macroeconomic regime changes, aggregate Focus statistics, and historical data revisions.
- Identify questions for further analysis.

References

- Banco Central do Brasil: Focus expectations data and 12-month forecast methodology.
- Instituto Brasileiro de Geografia e Estatistica: SIDRA table 1737, annual IPCA table, and IPCA documentation.
- Add the methodological reference used for the moving-block bootstrap.

A Additional Methods and AI Use

- Add any detailed data checks and simulation diagnostics needed beyond the main article.
- Explain how AI was used in research, coding, and writing, and how the results were checked.

B Implementation

- Include key R code or point to the supplementary analysis notebooks.
- Record the Focus snapshot, IBGE source, filters, package versions, and simulation seeds.